THE SUBOBJECT GENERATED BY TERMS OF DEGREE AT MOST N, AND VANISHING OF ITS NORMALIZED PART ABOVE N

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the factorization [m] -> [m'] -> [i] with [m] -> [m'] surjective and m' < m (source lines 2345-2352) is the point where m > n >= i is used; the splitting input is card L46157 (lemma-splitting-abelian-category); no verification is omitted.

Editorial note. The label lemma-simplicial-abelian-n-skel-sub names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Images under the simplicial operators. Statement lines 2310-2318. U'_m = sum of the images Im(U(phi)) over phi : [m] -> [i] with i <= n; the claimed sub simplicial object agrees with U in degrees <= n.
- The splitting of simplicial objects in abelian categories. Same chapter, \label{lemma-splitting-abelian-category} (card L46157). Proof lines 2332-2340: U is split, so it suffices to test the intersection with the normalized subobject against the sum of images of the N(U_{m'}).

Source reference. Original reference 017V: sources/stacks-simplicial/pinned-original-source.tex, source lines 2245–2245. Prerequisite pointer invoked at proof line(s) 2333; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object of $\mathcal{A}$. Let $n \geq 0$ be an integer. The rule*

$$U'_m = \sum\nolimits_{\varphi:[m]\to[i],\ i\leq n} \mathrm{Im}(U(\varphi))$$

defines a sub simplicial object $U' \subset U$ with $U'_i = U_i$ for $i \leq n$. Moreover, $N(U'_m) = 0$ for all $m > n$.

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Proof. Pick m, $i \leq n$ and some $\varphi : [m] \to [i]$. The image under $U(\psi)$ of $\mathrm{Im}(U(\varphi))$ for any $\psi : [m'] \to [m]$ is equal to the image of $U(\varphi \circ \psi)$ and $\varphi \circ \psi : [m'] \to [i]$. Hence U' is a simplicial object. Pick $m > n$. We have to show $N(U'_m) = 0$. By definition of $N(U_m)$ and $N(U'_m)$ we have $N(U'_m) = U'_m \cap N(U_m)$ (intersection of subobjects). Since U is split by Lemma 017V, it suffices to show that U'_m is contained in the sum

$$\sum\nolimits_{\varphi:[m]\to[m'] \text{ surjective, } m'<m} \mathrm{Im}(U(\varphi)|_{N(U_{m'})}).$$

By the splitting each $U_{m'}$ is the sum of images of $N(U_{m''})$ via $U(\psi)$ for surjective maps $\psi : [m'] \to [m'']$. Hence the displayed sum above is the same as

$$\sum\nolimits_{\varphi:[m]\to[m'] \text{ surjective, } m'<m} \mathrm{Im}(U(\varphi)).$$

Clearly U'_m is contained in this by the simple fact that any $\varphi : [m] \to [i]$, $i \leq n$ occurring in the definition of U'_m may be factored as $[m] \to [m'] \to [i]$ with $[m] \to [m']$ surjective and $m' < m$ as in the last displayed sum above. $\square$